Term 1. Today my requirement is to display the followings

```
1 - - - - > 1
            2
            3
2 - - - - > 1
            2
            3
3 - - - - > 1
            2
            3
4 - - - - > 1
            2
            3
5 - - - - > 1
            2
            3
```

Case ① for in for

```
for i in range(1,6): #outerloop
    print(i)
    for j in range(1,4): #inner
        print(j)
```

Case ② while in while

```
i=1
while (i<=5): # outer loop
    print(i)
    J=1
    while (J<=3): # inner loop
        print(J)
        J = J+1
    else:
        i=i+1
else:
```

(5,0,-1)

Example ①

```
for i in range (1,6):
    print ("value of i outer loop : {}".format(i))
    print ("— — — — ")
    for j in range (1,4):
        print (\t value of J-inner loop:{}".format(J))
    else:
        print("\tI am coming out of inner loop")
        print("— — — — ")
else:
    print ("I am coming out of outer loop").
```

for x in lst:
    print("\t {}".format(x)).

Logic). • Perfect Number: : Sum of factors of Given Number.

Given number = 6

Factors of 6 = 1, 2, 3

sum of factors = 1+2+3 = 6.

Hence, 6 is perfect.

2|8
2|4
2|2
 1

8 = 2+2+2+1
= 7 not perfect

• Armstrong Number: Sum of cubes of digits of Given number

Example 153 ---> 1**3 + 5**3 + 3**3.

= 1 + 125 + 27

= 153 (Given number).

Hence 153 is Armstrong

123 ---> 1**3 + 2**3 + 3**3

= 1 + 8 + 27

= 36. (!= Given number 123).

In [1]:

```
#examples 1:-
for i in range(1,6):
    print("value of i outer loop:{}".format(i))
    print("================================================")
    for j in range(1,4):
        print("\t value of j-inner loop:{}".format(j))
    else:
        print("\t i am coming out of inner loop")
        print("=======================================================")
else:
    print("i am coming out of outer loop")
```

```
value of i outer loop:1
=================================================
         value of j-inner loop:1
         value of j-inner loop:2
         value of j-inner loop:3
         i am coming out of inner loop
=====================================================
value of i outer loop:2
=================================================
         value of j-inner loop:1
         value of j-inner loop:2
         value of j-inner loop:3
         i am coming out of inner loop
=====================================================
value of i outer loop:3
=================================================
         value of j-inner loop:1
         value of j-inner loop:2
         value of j-inner loop:3
         i am coming out of inner loop
=====================================================
value of i outer loop:4
=================================================
         value of j-inner loop:1
         value of j-inner loop:2
         value of j-inner loop:3
         i am coming out of inner loop
=====================================================
value of i outer loop:5
=================================================
         value of j-inner loop:1
         value of j-inner loop:2
         value of j-inner loop:3
         i am coming out of inner loop
=====================================================
i am coming out of outer loop
```

In [1]:

```
#examples 2:
i=1
while (i<=5):  #outer loop
    print("value of i outer loop:{}".format(i))
    print("==========================================")
    j=1
    while(j<=3):  #inner loop
        print("\t value of j inner loop :{}".format(j))
        j=j+1
    else:
        i=i+1
        print("\t i am coming out of inner loop")
        print("==========================================")
else:
    print("i am coming out of of outer loop")
```

```
value of i outer loop:1
==========================================
         value of j inner loop :1
         value of j inner loop :2
         value of j inner loop :3
         i am coming out of inner loop
==========================================
value of i outer loop:2
==========================================
         value of j inner loop :1
         value of j inner loop :2
         value of j inner loop :3
         i am coming out of inner loop
==========================================
value of i outer loop:3
==========================================
         value of j inner loop :1
         value of j inner loop :2
         value of j inner loop :3
         i am coming out of inner loop
==========================================
value of i outer loop:4
==========================================
         value of j inner loop :1
         value of j inner loop :2
         value of j inner loop :3
         i am coming out of inner loop
==========================================
value of i outer loop:5
==========================================
         value of j inner loop :1
         value of j inner loop :2
         value of j inner loop :3
         i am coming out of inner loop
==========================================
i am coming out of of outer loop
```

In [3]:

```
#ex3:
i=1
while (i<=5):  #outer loop
    print("value of i outer loop:{}".format(i))
    print("==========================================")
    for j in range(3,0,-1):   #inner loop
        print("\t value of j inner loop :{}".format(j))
    else:
        i=i+1
        print("\t i am coming out of inner loop")
        print("==========================================")
else:
    print("i am coming out of of outer loop")
```

```
value of i outer loop:1
==========================================
         value of j inner loop :3
         value of j inner loop :2
         value of j inner loop :1
         i am coming out of inner loop
==========================================
value of i outer loop:2
==========================================
         value of j inner loop :3
         value of j inner loop :2
         value of j inner loop :1
         i am coming out of inner loop
==========================================
value of i outer loop:3
==========================================
         value of j inner loop :3
         value of j inner loop :2
         value of j inner loop :1
         i am coming out of inner loop
==========================================
value of i outer loop:4
==========================================
         value of j inner loop :3
         value of j inner loop :2
         value of j inner loop :1
         i am coming out of inner loop
==========================================
value of i outer loop:5
==========================================
         value of j inner loop :3
         value of j inner loop :2
         value of j inner loop :1
         i am coming out of inner loop
==========================================
i am coming out of of outer loop
```

In [4]:

```
#ex4:
for i in range(5,0,-1):
    print("value of i outer loop:{}".format(i))
    print("=================================================")
    j=3
    while(j>0):
        print("\t value of j inner-loop :{}".format(j))
        j=j-1
    else:
        i=i+1
        print("\t i am coming out of inner loop")
        print("==========================================")
else:
    print("i am coming out of of outer loop")
```

```
value of i outer loop:5
=================================================
         value of j inner-loop :3
         value of j inner-loop :2
         value of j inner-loop :1
         i am coming out of inner loop
==========================================
value of i outer loop:4
=================================================
         value of j inner-loop :3
         value of j inner-loop :2
         value of j inner-loop :1
         i am coming out of inner loop
==========================================
value of i outer loop:3
=================================================
         value of j inner-loop :3
         value of j inner-loop :2
         value of j inner-loop :1
         i am coming out of inner loop
==========================================
value of i outer loop:2
=================================================
         value of j inner-loop :3
         value of j inner-loop :2
         value of j inner-loop :1
         i am coming out of inner loop
==========================================
value of i outer loop:1
=================================================
         value of j inner-loop :3
         value of j inner-loop :2
         value of j inner-loop :1
         i am coming out of inner loop
==========================================
i am coming out of of outer loop
```

In [7]:

```
#inner loops multiplication table:
n= int(input("how many mul table  do you want:"))
if (n<=0):
    print("{} is invalid input:".format(n))
else:
    for num in range(1,n+1):   #outer loop
        print("==========================================")
        print("multable for {}".format(num))
        print("==========================================")
        for i in range(1,11):     #inner loop
            print("\t {}*{}={}".format(num,i,num*i))
        else:
            print("=======================================")
```

```
==========================================
multable for 1
==========================================
         1*1=1
         1*2=2
         1*3=3
         1*4=4
         1*5=5
         1*6=6
         1*7=7
         1*8=8
         1*9=9
         1*10=10
=======================================
==========================================
multable for 2
==========================================
         2*1=2
         2*2=4
         2*3=6
         2*4=8
         2*5=10
         2*6=12
         2*7=14
         2*8=16
         2*9=18
         2*10=20
=======================================
==========================================
multable for 3
==========================================
         3*1=3
         3*2=6
         3*3=9
         3*4=12
         3*5=15
         3*6=18
         3*7=21
         3*8=24
         3*9=27
         3*10=30
=======================================
==========================================
multable for 4
==========================================
         4*1=4
         4*2=8
         4*3=12
         4*4=16
         4*5=20
         4*6=24
         4*7=28
         4*8=32
         4*9=36
         4*10=40
=======================================
```

In [8]:

```
#random multiplication number:
n= int(input("how many mul table  do you want:"))
if (n<=0):
```

```
    print("{} is invalid input:".format(n))
else:
    lst=list()
    for i in range (1,n+1):
        val=int(input("enter {} number:".format(i)))
        lst.append(val)
    else:
        print("===================================================")
        print("given list of values :{}".format(lst))
        print("====================================================")
        for num in lst:
            if(num<=0):
                print("{} invalid input".format(num))
            else:
                print("================================================")
                print("multable for {}".format(num))
                print("=================================================")
                for i in range(1,11):
                    print("\t {}*{}={}".format(num,i,num*i))
                else:
                    print("==============================================")
```

```
===================================================
given list of values :[3, 4]
====================================================
================================================
multable for 3
=================================================
         3*1=3
         3*2=6
         3*3=9
         3*4=12
         3*5=15
         3*6=18
         3*7=21
         3*8=24
         3*9=27
         3*10=30
==============================================
================================================
multable for 4
=================================================
         4*1=4
         4*2=8
         4*3=12
         4*4=16
         4*5=20
         4*6=24
         4*7=28
         4*8=32
         4*9=36
         4*10=40
==============================================
```

In [10]:

```
#inner loops primes:
n= int(input("enter the range of prime numbers:"))
if (n<=1):
    print("{} is invalid".format(n))
else:
    lst=list()
```

```
    for num in range(2,n+1):
        res="prime"
        for i in range(2,num):
            if(num%i==0):
                res="not prime"
                break
        if(res=="prime"):
            lst.append(num)
    else:
        print("list of primes")
        for v in lst:
            print("\t{}".format(v))
```

```
list of primes
        2
        3
        5
        7
```

In [ ]: